HTTP IN DETAIL

TryHackMe Learning & Practical Report

Prepared by	Belal Rabea Khalifa
Program	IqraSec Academy, WAPT Internship (Cohort 02), Phase 1
Task	1. Web Fundamentals
Platform	TryHackMe
Room	HTTP in Detail (https://tryhackme.com/room/httpindetail)
Focus	HTTP, requests and responses, methods, status codes, headers, cookies
Completion	100% (all 7 tasks)
Date	October 2026

1. Objective

Learn how HTTP works (requests, responses, methods, status codes, headers, cookies) and practice sending HTTP requests in the room's emulator.

2. Concept / Understanding

- **HTTP** (HyperText Transfer Protocol) is a stateless application layer protocol. It runs on top of TCP and is used to transfer resources like web app data.
- HTTP is the common language between web browsers (Chrome, Edge, Firefox) and web servers (Apache, Nginx, IIS). Because of it, any browser can talk to any server.
- The browser sends an **HTTP request** to the server, and the server sends back an **HTTP response**. HTTPS is the same thing but encrypted.
- A request has a request line (method, path, version), headers, and a body. A response has a status line, headers, and a body.
- **Methods:** GET reads data, POST sends data, PUT updates data, DELETE removes data, HEAD returns the headers only.
- **Status codes:** 2xx success (200 OK), 3xx redirect (301), 4xx client error (403 Forbidden, 404 Not Found), 5xx server error (500).
- **Headers** carry extra information about the request or response, for example Host, User-Agent, Content-Type, Cookie, Set-Cookie.
- **Cookies:** HTTP is stateless, so the server does not remember the user. It sends a cookie (Set-Cookie) and the browser sends it back (Cookie) on the next requests. This is how sessions work.

Terms I noted:

Term	Meaning
client	the browser
version	HTTP/1.1
type	HTTP or HTTPS
\r\n	new line
body	the data
URL	the location
resource	the file

curl options I noted:

Option	Use
curl -X	choose the method (GET, POST, PUT...)
curl -H	set a header (like User-Agent)
curl -I	request the page with the HEAD method
curl -d	send data

3. Practical Work

I finished all 7 tasks of the room (What is HTTP(S), Requests and Responses, HTTP Methods, HTTP Status Codes, Headers, Cookies, Making Requests). In Task 7, I used the room's HTTP request emulator and sent these 5 requests:

#	Request	Details
1	GET /room	basic request
2	GET /blog	URI parameter id=1
3	DELETE /user/1	delete a user
4	PUT /user/2	body parameter username=admin
5	POST /login	body parameters username=thm, password=letmein

4. Tools Used

- TryHackMe room: HTTP in Detail
- HTTP request emulator (Task 7)

5. Evidence

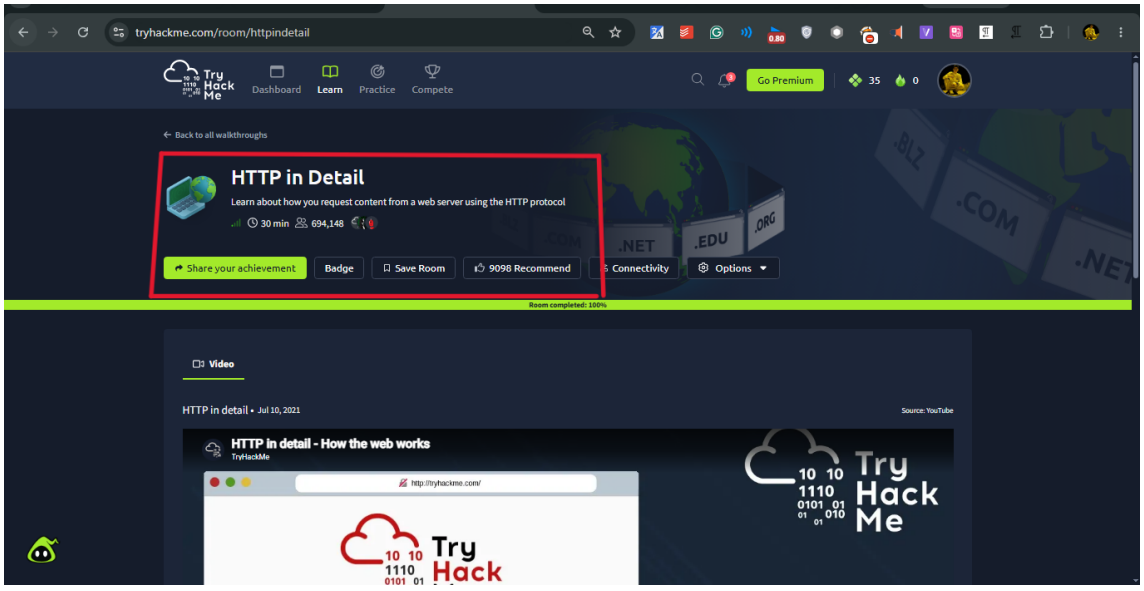


Figure 1: HTTP in Detail room page, 100% completed.

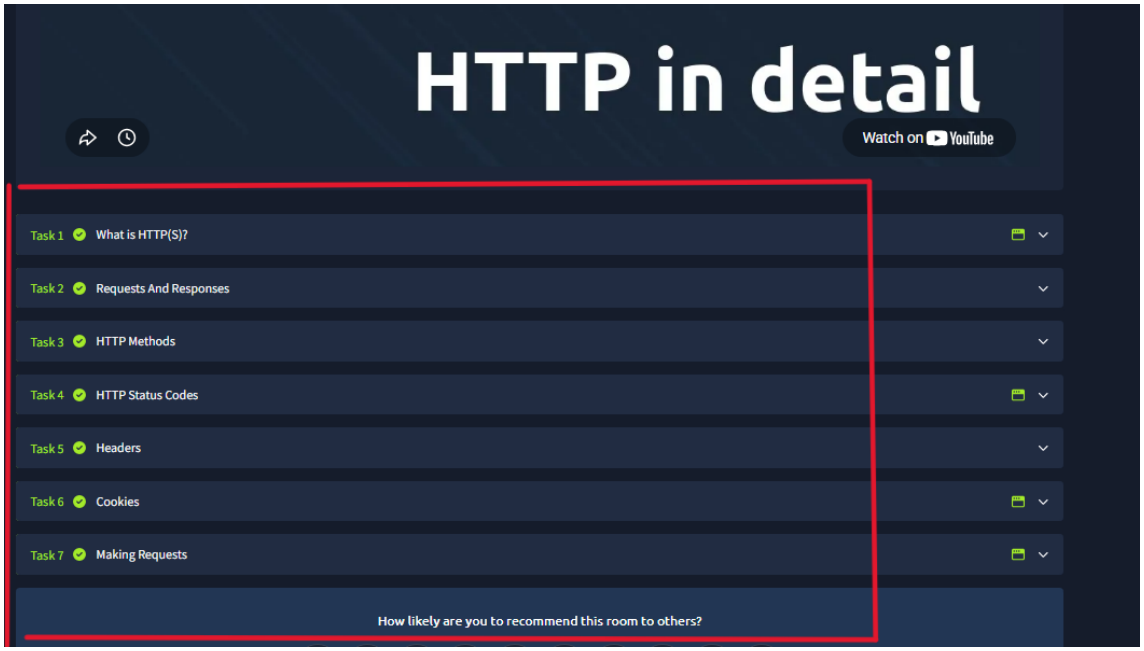


Figure 2: all 7 tasks of the room are completed.

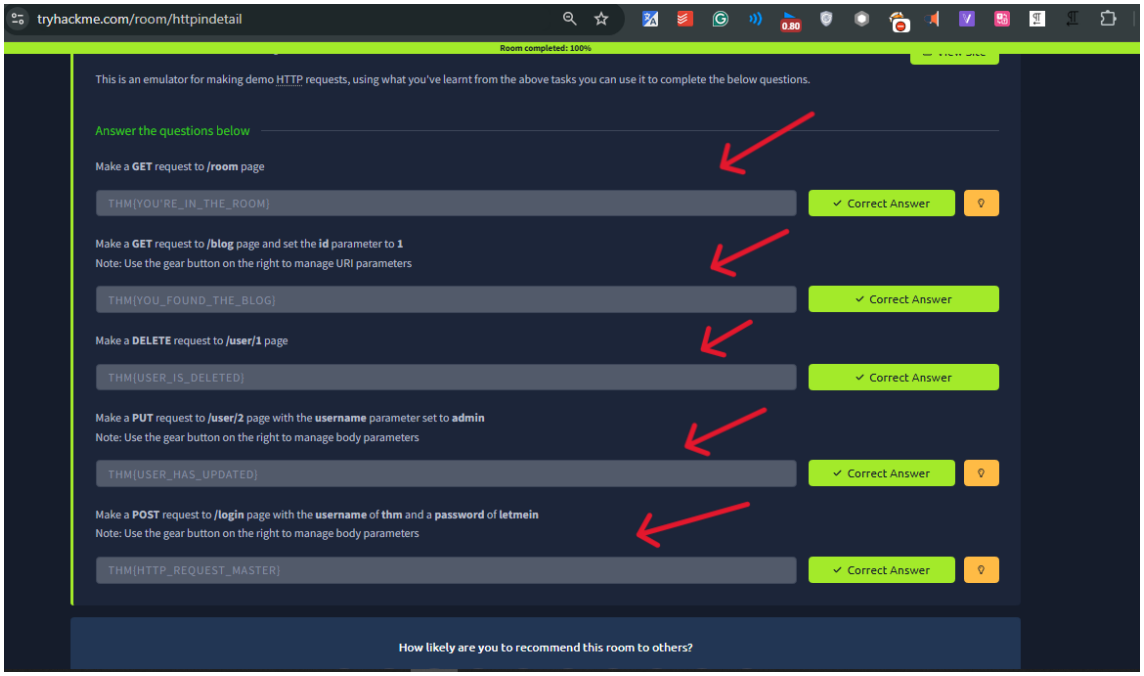


Figure 3: the 5 emulator requests, all with correct answers.

6. Result

Completed the room 100%. All 5 emulator requests (GET, GET with URI parameter, DELETE, PUT, POST) gave the correct answers.

7. Key Learnings

- HTTP is the common language between browsers and servers, and it runs on top of TCP.
- HTTP is stateless, so cookies are used to keep a user's session.
- Each method has a job: GET reads, POST sends, PUT updates, DELETE removes.
- Parameters can go in the URL (like id=1) or in the request body (like username and password).